Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/10/2008
PI Name:	Zinchenko, Alexander Ph.D.		
Project Title:	Multiphase Flow of Emulsions Through Porous Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	05/31/2007
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 05/31/2007 per PI (12/06) NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis , Robert H. (University of Colorado at Boulder)		
Grant/Contract No.:	NNC05GA55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Multiphase emulsion and bubbly flows through packed beds and other porous media are important in space-based applications such as liquid and solid waste recycling and food and fuel processing. Dynamic simulation capabilities will be established to describe multiphase flows of concentrated emulsions through porous media and packed beds in the presence and absence of gravitational effects, with supporting experiments on selected test problems. The specific tasks include: (i) Development of new simulation methods for deformable drops and rigid particles - Highly efficient computer algorithms based on hybrid boundary-integral-multipole techniques will be developed to simulate flows of emulsions of deformable drops (or bubbles) through porous media and fixed beds of rigid particles with prescribed microstructure. (ii) Development of new simulation methods for spherical drops and rigid particles - For small drops or bubbles that exhibit little deformation, a purely multipole algorithm will be constructed using geometry perturbation to handle near-contact interactions and lubrication forces. Conditions when the drops or bubbles become trapped within the bottlenecks of the porous medium will be identified. (iii) Development of new methods for predicting the microstructure of packed beds - Using a novel algorithm based on random placements and mechanical equilibrium, random close and loose packings of sperical particles will be generated and used in hydrodynamical simulations of multiphase flows through these packed beds. (iv) Performance of ground-based test experiments - A focused set of microphysical experiments on single-drop interactions with one, two, or three solid particles will be performed to test and guide some of the most sensitive features of the simulations. The effects of gravity will be assessed by varying the ratio of drop and matrix densities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We studied flow-induced squeezing of a deformable drop through tight three-dimensional constrictions formed by several solid particles rigidly held in a uniform flow. The problem is a close prototype of drop-solid interactions for emulsion flow through a granular material. The effects of the constriction type (2-particle or 3-particle), capillary number, and viscosity ratio on the drop velocity in the throat, exit time and drop-solid spacing have been explored; critical capillary numbers (for squeezing to occur) were determined. The drop non-deformed diameter is from two to several times larger than the inner constriction diameter and the drop can decelerate up to 10(3)-10(4) times before being able to pass through. The novel boundary-integral methodology developed for this problem was further incorporated into a large-scale multipole-accelerated multiparticle- multidrop algorithm suitable for simulating an emulsion flow through dense granular materials. Flow of a periodic emulsion squeezing though a 50% cubic lattice of particles was studied in detail, including pressure gradient-flow rate relationships for both phases and critical capillary numbers. Drops are found to move, on the average, faster than the continuous phase (a finite size ratio effect) in a wide range of conditions, except for very close to critical where this trend would be reversed. Multidrop squeezing through a random granular material showed both trends: drops moving slower than the fluid for high viscosity contrast, and faster for the homoviscous case. Statistical steady state is reached with drops passing the constrictions 1-2 times only. The ability of drops to break in this geometry is severely hampered by solid particles . Axisymmetrical single drop motion through a toroidal constriction was studied, to delineate the distinctions between the gravity-induced and flow-induced squeezing mechanisms and find universal near-critical scaling for the squeezing time. A related problem of a deformable drop driven by Poiseuille flow between two plates was considered; the drop velocity, transient and steady-state, was calculated for small and tightly fitting drops. In another related problem of a deformable drop sliding down an inclined wall, including the case of close contact, we found that the steady-state gap is a decreasing function of the viscosity ratio, and drop velocity is generally a non-monotonic function of the Bond number, in good quantitative agreement with our experiments.
Bibliography Type:	Description: (Last Updated: 01/10/2008)
Articles in Peer-reviewed Journals	Zinchenko AZ, Davis RH. "Algorithm for direct numerical simulation of emulsion flow through a granular material." Journal of Computational Physics , submitted May 2007. , May-2007
Articles in Peer-reviewed Journals	Griggs AJ, Zinchenko AZ, Davis RH. "Gravity-driven motion of a deformable drop or bubble near an inclined plane at low Reynolds number." International Journal of Multiphase Flow, In Press, Corrected Proof, Available online 30 October 2007. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2007.10.006 , Oct-2007
Articles in Peer-reviewed Journals	Griggs AJ, Zinchenko AZ, Davis RH. "Low-Reynolds-number motion of a deformable drop between two parallel plane walls." International Journal of Multiphase Flow. 2007 Feb;33(2):182-206. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2006.06.012 , Feb-2007
Articles in Peer-reviewed Journals	Zinchenko AZ, Davis RH. "A boundary-integral study of a drop squeezing through interparticle constrictions." J Fluid Mech. 2006;564:227-66. http://dx.doi.org/10.1017/S0022112006001479 , Mar-2006
Articles in Peer-reviewed Journals	Zinchenko AZ, Davis RH. "Squeezing of a periodic emulsion of deformable drops through a cubic lattice of spheres." Physics of Fluids, submitted May 2007. , May-2007
Articles in Peer-reviewed Journals	Rother MA, Zinchenko AZ, Davis RH. "Surfactant effects on buoyancy-driven viscous interactions of deformable drops." Colloids and Surfaces A: Physicochemical and Engineering Aspects 2006 Jul 20;282-283:50-60. http://dx.doi.org/10.1016/j.colsurfa.2005.11.086 , Jul-2006
Dissertations and Theses	Griggs AJ. "The low-Reynolds number motion of drops and bubbles near solid planar surfaces." Ph.D. thesis, University of Colorado, 2007. , Jun-2007